

Lab B — Microphone scattering

34870 Electroacoustics

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Laboratory date: Tuesday 22 September 2026, 10:00–12:00 (building 354, rooms 028/025)

Working draft. Measured on 22 September 2026; this skeleton follows the official E26 lab brief. Replace the coloured prompts and figure boxes with your group’s evidence. Given settings and theory are not measured results. Keep raw data unchanged and record which files were analysed.

1 Purpose and theoretical basis

The experiment investigates how a microphone-shaped rigid body changes the pressure at its front face. A wooden, enlarged microphone model makes this scattering measurable using a small microphone. The angle-dependent response is compared with a boundary element method (BEM) calculation and with microphone handbook free-field corrections.

For a body radius a , the dimensionless frequency is

$$ka = \frac{2\pi fa}{c}, \quad a = \frac{D_{\text{mock}}}{2}. \quad (1)$$

Here c is the sound speed. Similar geometries are compared at equal ka . The measured voltage ratio and the normalized scattering response are

$$H_{\theta}(f) = \frac{V_{\text{mic},\theta}(f)}{V_{\text{ref},\theta}(f)}, \quad H_0^{\text{ref}}(f) = \frac{V_{\text{mic,no mock}}(f)}{V_{\text{ref,no mock}}(f)}, \quad (2)$$

$$H_{\text{norm},\theta}(f) = \frac{H_{\theta}(f)}{H_0^{\text{ref}}(f)}, \quad \Delta L_{\theta}(f) = 20 \log_{10} |H_{\text{norm},\theta}(f)|. \quad (3)$$

The subscript 0 in H_0^{ref} denotes the reference run, not zero-degree incidence. Normalization removes the common measurement-chain response only when its settings and the loudspeaker and UMIK positions remain unchanged.

To complete: Explain the low- ka limit, the physical origin of the angle dependence, and the conditions under which the spherical loudspeaker wave approximates a plane wave.

2 Equipment and geometry

Item	Recorded identity or setting
PC / internal sound card	Lab desktop, internal card (Line In reference, Speakers output): <i>[fill in]</i>
UMIK MiniDSP / USB microphone	S/N (7080329, 7080332 or 7080335): <i>[fill in]</i>
Loudspeaker / amplifier	Type and gain/volume: <i>[fill in]</i>
Wooden microphone mock-up	Diameter D_{mock} : <i>[fill in]</i> ; length: <i>[fill in]</i>
Mock-up back end	Round / flat: <i>[fill in]</i>
Loudspeaker–UMIK distance	$r =$ <i>[fill in]</i> ; driver alignment: <i>[fill in]</i>
UMIK tip–front–face gap	<i>[fill in]</i> ; centring and height: <i>[fill in]</i>
Angle reference / temperature	<i>[fill in]</i>

To complete: Transfer actual geometry from LabB/data/labB_log.md; do not replace it with BEM nominal dimensions.

3 Part 0: Anechoic chamber and free-field check

The brief specifies a chamber cutoff frequency of 125 Hz. The expected pressure magnitude of an approximately spherical wave varies as $1/r$, giving

$$\Delta L_{21} = 20 \log_{10} \left(\frac{|p(r_2)|}{|p(r_1)|} \right) \simeq 20 \log_{10} \left(\frac{r_1}{r_2} \right). \quad (4)$$

If performed: carry out the distance check without the mock-up, before fixing the source and microphone positions for Part 1. If omitted, state this explicitly.

Run	Distance / m	Data file	Observed difference / dB
Reference distance	<i>[fill in]</i>	labB_dist93cm.mat	Reference
Second distance	<i>[fill in]</i>	labB_dist140cm.mat	<i>[fill in]</i>
Third	<i>[fill in]</i>	labB_dist280cm.mat	<i>[fill in]</i>

Figure to add

Measured distance ratios and the $1/r$ prediction; mark 125 Hz.
labB_free_field_check.png

To complete: Check the distances against the log (the lab-PC file `no_mockup_1.8m` is suspected to be a typo: does the level difference agree with 1.8 or 2.8 m?). Note that Part 0 was measured *after* the series. State whether the distance test supports free-field conditions over the useful band. Discuss near-field behaviour, directivity and the chamber cutoff where relevant.

4 Part 1: Scattering measurement procedure

1. Connect the UMIK before starting MATLAB and check the selected devices. Begin with low amplifier output and inspect the measurement spectra for noise and clipping.
2. After any distance tests, fix the loudspeaker and upward-pointing UMIK. Measure and save a reference without the mock-up.
3. Position the mock-up front-face centre almost touching the UMIK tip. Rotate only the mock-up and measure at least 0° , 45° and 90° .
4. Inspect and save every run. A repeated no-mock-up reference at the end is a useful additional drift check. Copy and verify the data before removing files from the lab PC.

Setup sketch to add: sound-card output to line-in reference and amplifier; amplifier to loudspeaker; UMIK to PC through USB. Label the fixed positions, face gap and mock-up incidence angle. See the brief's Figure 1.

Parameter	Prescribed value	Parameter	Prescribed value
f_1	50 Hz	f_2	10 000 Hz
Tones/octave	24	Resolution	1 Hz
Averages N_{av}	32	UMIK incidence	90°

The UMIK incidence selects its correction and is *not* the mock-up angle.

To complete: Record the actual settings and any departures from the prescribed values.

4.1 Measurement record and quality checks

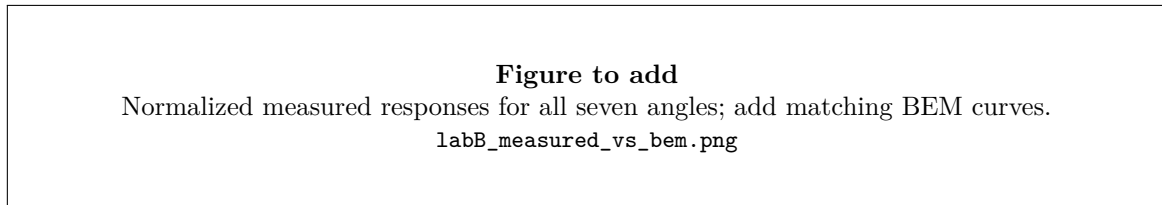
Condition	File / repeat used	Angle	Quality / notes
Trial	test.mat	—	[fill in]
Mock-up	labB_ang000.mat	0°	[fill in]
Mock-up	labB_ang015.mat	15°	[fill in]
Mock-up	labB_ang030.mat	30°	[fill in]
Mock-up	labB_ang045.mat	45°	[fill in]
Mock-up	labB_ang060.mat	60°	[fill in]
Mock-up	labB_ang075.mat	75°	[fill in]
Mock-up	labB_ang090.mat	90°	[fill in]
No mock-up	labB_nomockup.mat	—	[fill in]
End reference	[fill in]	—	[fill in]

Lab-PC files are in `data/LabMastergroup10/`; the `labB_*.mat` names are the copies written by `import_group_files.m`.

To complete: Record the order in which the angles were measured and when the reference was taken (before or after the series?). The course script `labB.m` saved only `h`: say what that means for the noise and distortion check. Comment on the separation between tones and the background in the full spectra, clipping/distortion, and any repositioning.

4.2 Normalized angle responses

Calculate $H = \text{specn}(:, 2) ./ \text{specn}(:, 1)$ for every run, then divide by the no-mock-up reference on the same frequency grid. Plot ΔL_θ with a logarithmic frequency axis. Retain the raw full spectra for noise and distortion checks.



Incidence	Low-frequency level / dB	Feature frequency / Hz	Feature level / dB
0°	[fill in]	[fill in]	[fill in]
15°	[fill in]	[fill in]	[fill in]
30°	[fill in]	[fill in]	[fill in]
45°	[fill in]	[fill in]	[fill in]
60°	[fill in]	[fill in]	[fill in]
75°	[fill in]	[fill in]	[fill in]
90°	[fill in]	[fill in]	[fill in]

To complete: Identify the useful band and describe peaks/notches and angle dependence using actual values from your curves. Distinguish centre-point pressure from diaphragm-averaged pressure.

4.3 Reference stability (additional quality check)

If the end reference exists, evaluate $\Delta L_{\text{drift}} = 20 \log_{10} |H_{\text{end}}/H_{\text{start}}|$. Insert `labB_reference_drift.png` and report the band and size of any drift.

To complete: State whether the reference remained stable enough for normalization; if no end reference was taken, describe that limitation instead of reporting zero drift.

5 Part 2: BEM simulation and spatial effects

The supplied `CylinderPlaneWave` model assumes plane-wave incidence. Its frequency grid is 50 Hz to 5382 Hz in 1/12-octave steps. `AngleS` is in radians; `CylEnd` selects a round or flat back end. Compare measurements and simulation only over their common, reliable band.

Output / choice	Comparison to make
<code>p_centre</code>	Pressure at the front-face centre; compare with normalized UMIK data.
<code>p_FP</code>	Field point near the face; default is 3 cm in front. Document the coordinates used and relate them to the measured tip gap.
<code>p_avg</code>	Parabolically weighted face average; discuss a real diaphragm's spatial averaging.
Round vs. Flat	Compare rear-end geometry while holding incidence fixed.

Figure to add

Centre, field point, weighted average and flat/round back at matched incidence.

`labB_bem_positions.png`

To complete: An existing BEM prediction is in the figures folder. Verify its geometry, field-point coordinates and model settings before inserting it. Label it as simulation, not measurement.

5.1 Where was the UMIK tip?

The field point [`r_FP``z_FP`] is in the cylinder's own coordinates (face at  $z = 0$ ), so  $z_{FP}$  is the distance from the face to the microphone. `LabB/bem/explore_gap.m` sweeps it and overlays the measured curve at the same angle.

Figure to add

BEM at several gaps in front of the face, with the measured 0° curve on top.

`labB_gap_sweep.png`

To complete: Which gap reproduces the measurement? Explain the notch physically (incident and reflected waves in front of the face) and compare with the gap recorded in the log. Check whether a simple quarter-wave estimate $c/(4z)$ gives the same answer as the BEM.

5.2 Agreement and limitations

Comparison	Observation / quantitative difference	Possible explanation
Measured vs. face centre	<i>[fill in]</i>	<i>[fill in]</i>
Measured vs. field point	<i>[fill in]</i>	<i>[fill in]</i>
Centre vs. average	<i>[fill in]</i>	<i>[fill in]</i>
Round vs. flat rear	<i>[fill in]</i>	<i>[fill in]</i>

To complete: Discuss finite source distance, tip position, supports and UMIK body scattering, geometry mismatch, and numerical resolution. Do not extrapolate the BEM result beyond its available frequency grid. State which pressure definition best represents the experiment.

6 Part 2 continued: Scale to real microphones

For a geometrically similar microphone of diameter D_{mic} , define

$$s = \frac{D_{\text{mock}}}{D_{\text{mic}}}, \quad f_{\text{mic}} = s f_{\text{mock}}. \quad (5)$$

The smaller microphone shows corresponding features at *higher* frequencies. Conversely, a microphone datasheet frequency maps to the mock-up axis as $f_{\text{mock}} = f_{\text{mic}}/s$. The normalized pressure level is unchanged under ideal geometric scaling with the same medium.

Nominal size	Diameter used / mm	Scale factor s	Handbook type / grid condition
1"	[fill in]	[fill in]	[fill in]
1/2"	[fill in]	[fill in]	[fill in]
1/4"	[fill in]	[fill in]	[fill in]
1/8"	[fill in]	[fill in]	[fill in]

To complete: Use the measured mock-up diameter and state the physical diameter convention for each real microphone. Nominal inch size and actual body diameter need not be identical.

Figure to add
Scaled measured and BEM curves, compared with the handbook corrections.
`labB_scaled_to_microphones.png`

To complete: For each size, compare levels and characteristic frequencies at matching incidence. Identify the handbook figure and whether the protection grid is fitted. The current processing figure contains scaled measurements; add scaled BEM/datasheet evidence as needed.

7 Uncertainty and conclusion

Sources to assess: repeatability, angle reading, tip gap/centring, diameter, reference drift, noise, sound speed and deviations from plane-wave/rigid-body assumptions. For $s = D_{\text{mock}}/D_{\text{mic}}$, independent diameter uncertainties give

$$\frac{u(s)}{s} = \sqrt{\left(\frac{u(D_{\text{mock}})}{D_{\text{mock}}}\right)^2 + \left(\frac{u(D_{\text{mic}})}{D_{\text{mic}}}\right)^2}. \quad (6)$$

To complete: Rank the main limitations using your evidence. For any numerical uncertainty, state the input values, distributions and coverage; account for shared-reference correlation rather than assuming all normalized curves are independent.

To complete: Conclude with the measured angle dependence, BEM agreement and validity of the scaled comparison. Give the frequency band over which those conclusions are supported.

Sources and processing record

DTU 34870, *Lab B – Microphone scattering – E26*, 34870_Lab_B_CylinderScattering_E2026.pdf: pp. 1–5 (procedure, equations and BEM), followed by the microphone handbook excerpts. Source: course vault / DTU Learn.

Repository workflow: raw lab-PC files in LabB/data/LabBmastergroup10/ (unchanged), LabB/matlab/import_group_files.m (rebuilds the frequency axis, writes labB_*.mat), LabB/bem/run_bem.m, LabB/bem/explore_gap.m, LabB/matlab/process_labB.m. Update D_mockup before processing. The BEM package is

obtained separately from DTU Learn. Course measurement files: the 21 September 2026 re-upload (34870 - Lab B (2).zip), which finds the internal sound card by the names **Line In** and **Speakers**.

Analysis version / selected repeats / deviations: *[fill in]*